

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

19CSE73 / 16CSE73 – SOCIAL AND WEB ANALYTICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

	Marks	BT*	CO*	PO*
1. a) What is social media analytics? List its benefits.	6	L*1	1	1
b) List the challenges faced in social media data collection.	6	L1	1	1
c) Explain the following built-in function of R with basic syntax and example. i) apply() ii) rbind() iii) format() iv) summary() v) str() vi) nlevels() vii) paste() viii) barplot()	8	L2	2	1
2. a) Illustrate the paradigm of web analytics 2.0 framework with diagram.	8	L2	1	1
b) Define the following terms. i) ROI ii) Brand reputation iii) Dashboard iv) Web marketing v) Customer sentiment analysis vi) Bounce rate	6	L1	1	1
c) List and explain the features of R.	6	L2	2	1
3. a) List six R objects and explain with an example.	12	L2	2	1
b) Explain the impacts of social media on business.	8	L2	1	1

Unit – II

4. a) What is KPI and Metric? Explain Various types of critical web metrics.	10	L1	3	1
b) Explain the following terms of twitter mining with necessary examples. i) Examining Patterns in retweets. ii) Computing the Lexical diversity of tweets.	10	L2	4	1
5. a) Why need to align social objectives with business goal? Explain. Assume that a web site has pages A through C and it has only one session per day that exists, with the following page view order as follows: Monday: Page A > Page C > Page B Tuesday: Page B > page A > Page C Wednesday: Page A > Page C > Page B Thursday: Page C > Exit Friday: Page C > Page B > Page A Calculate Bounce Rate and Exit Rate for each page.	10	L3	3	1

P.T.O.

- b) What is social graph API?
 Explain in detail with necessary examples the following terms of facebook mining.
 i) Analyzing the social graph connections.
 ii) Analyzing the Coke vs Pepsi facebook pages.

10 L2 4 1

6. a) Identify and explain the common social business objectives.
 b) Discuss how to create a Twitter API Connection and extract tweet entities.

10 L2 3 1

10 L2 4 1

Unit – III

7. a) Explain the most common data mining applications related to social networking sites.

10 L2 5 1

- b) Write a short note on following.

i) Transfer Learning in Heterogeneous Networks.

ii) Event map.

10 L1 5 1

8. a) What is classification and clustering in machine learning? Write the challenges arise in the text classification. Briefly explain the classification algorithms used for Text Mining in Social Networks.

10 L2 5 1

- b) Explain Graph Exploration by Index based graph exploration with respect to Keyword search over graph data.

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
